

Augmenting Trees so that Every Three Vertices Lie on a Cycle

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Abstract

The minimum number of edges that must be added to a tree so that every three vertices lie on a cycle is determined. The proof is constructive.

Key words: graph augmentation, 3-cyclable, tree, cycle

1 Introduction

In this paper we consider undirected graphs and use the notation of [16]. Let P be a property satisfied by the complete graph. The process of adding a minimum number of edges to a given graph G so that the resulting graph has property P is called *augmenting G to have property P* . The problem of augmenting a graph to be k -connected or k -edge-connected for some k has been studied by several people. In this paper we consider an augmentation problem which is a variation of those problems.

One of the first results on such augmentations is due to Eswaran and Tarjan [1] and Plesník [11]. They showed that the augmentation problem for general graphs to be 2-connected can be solved in polynomial time. Algorithms for trees were also found by Slater [12]. Eswaran and Tarjan [1] and Plesník [11] also showed that the augmentation problem for weighted graphs to be 2-connected is NP-complete. That is, given a vertex set V , a mapping from $\binom{V}{2}$ to the nonnegative integers and a bound B , it is NP-hard to decide if one can choose a set of edges of weight at most B which form a spanning 2-connected graph.

The problem of augmenting a given 2-connected graph to be 3-connected was solved by Watanabe and Nakamura [14] and a linear-time algorithm given by Hsu and Ramachandran [7]. Further results are due to Hsu [6] and Jordán [5]. However, the general problem of augmenting a graph to be k -connected remains open, even when restricted to trees.

There has been more success for the problem of augmenting a graph to have high edge-connectivity. Eswaran and Tarjan [1] and Plesník [11] developed an algorithm for augmenting a graph to be 2-edge-connected. The problem of augmenting a graph to be k -edge-connected was solved efficiently by Watanabe and Nakamura [13], Frank [2] and Naor, Gusfield and Martel [10].

There has also been success for the problem of augmenting a digraph to have high connectivity, including partial results by Eswaran and Tarjan [1] and Masuzawa et al. [8,9]. Recently, Frank and Jordán [3] gave for every k a polynomial-time algorithm for the problem of augmenting a directed graph to be k -connected.

Also, the problem of augmenting a tree to a hamiltonian graph has been solved; see for example Goodman and Hedetniemi [4].

In this paper we consider an augmentation problem which generalizes both the original Eswaran–Tarjan and Plesník problem and the hamiltonian augmentation problem. A graph is 2-connected if and only if every pair of vertices lie on a common cycle, while a graph is hamiltonian if and only if all the vertices lie on a common cycle. In a 3-connected graph every triple of vertices lie on a common cycle. But the converse is false: consider the graph which is itself a cycle. Thus we define:

A graph G satisfies the nVC property if every n vertices lie on a common cycle.

We focus here on the 3VC property. It is a weaker condition than 3-connectedness. Watkins and Mesner [15] characterized those graphs which satisfy 3VC. Their result is:

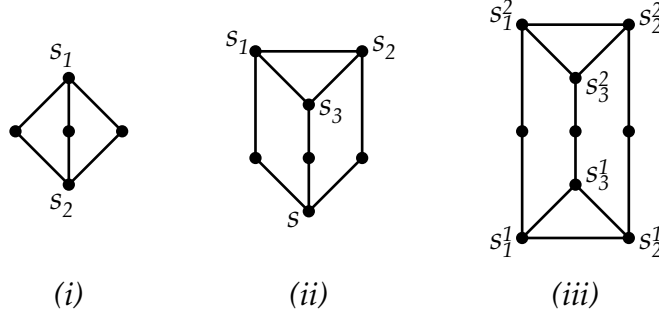


Fig. 1. 2-connected graphs not satisfying 3VC

Theorem 1 [15] *A graph G has every 3 vertices on a common cycle if and only if*

a) it is 2-connected and

b) it does not contain a set S with any of the following:

(i) $|S| = 2$ and $G - S$ has at least three components;

(ii) $S = \{s_1, s_2, s_3, s\}$ where each pair of elements of S are joined by a path in G having no interior vertex in S , and the three pairs $\{s_i, s\}$ are all cut-sets; or

(iii) $S = \{s_1^1, s_2^1, s_3^1, s_1^2, s_2^2, s_3^2\}$ where a pair of elements s_i^j and s_k^l are joined by a path in G having no interior vertex in S if and only if $i = k$ or $j = l$, and the three pairs $\{s_i^1, s_i^2\}$ are all cut-sets. See Figure 1 for some examples.

Watkins and Mesner [15] also showed that for $k \geq 3$, a k -connected graph has every $k + 1$ vertices on a common cycle if and only if the removal of k vertices does not yield $k + 1$ components. Of course, k -connectedness is not necessary. In fact any hamiltonian graph has the n VC property for $2 \leq n \leq |V(G)|$.

In this paper we are interested in augmenting a graph to have the 3VC property. Let $aug(G)$ be the minimum number of edges that must be added to G so that the resulting graph satisfies 3VC. We present here a method for computing $aug(T)$ when T is a tree, and for finding a minimum augmenting set.

2 A Lower Bound

We start with a discussion of a lower bound. Whether a vertex has degree less than 3 or at least 3 will play a key role. While our focus is on trees, the lower bound holds for all graphs.

For a graph G we define: $L(G)$ (or L if G is clear from context) as the set of vertices of degree 1 (the end-vertices), and $M(G)$ (or M if G is clear from context) as the set of vertices of degree 2. We define $l(G)$ as the number of

end-vertices of G . A leaf is an edge incident with an end-vertex. Equivalently, $l(G)$ is the number of leaves provided the order of G is at least 3.

We say that a vertex is *small* if it has degree at most 2, and *big* if it has degree at least 3. We say that a subset D of small vertices is *decent* if for every big vertex in G at most two of its small neighbours are not in D . We define $d(G)$ as the minimum cardinality of a decent set. For example, $d(G) = 0$ iff every vertex has at most two small neighbours.

The connection with the 3VC property is as follows.

Lemma 2 *If G satisfies 3VC then:*

- (a) *G has no end-vertex; and*
- (b) *the empty set is decent in G .*

PROOF. Part (a) is obvious since G is 2-connected. To prove (b) suppose that the empty set is not decent in G . That means there exists some vertex v with three small neighbours a , b and c . But these three small vertices cannot simultaneously be in a cycle as they each have degree 2 and have a common neighbour.

In other words, if G satisfies 3VC then $l(G) = d(G) = 0$. The converse is not true even if one prescribes 2-connectedness; consider for example the join $K_2 + 3K_m$, for $m \geq 2$.

Lemma 3 *The quantity $d(G)$ is computable in polynomial time.*

PROOF. Define a small vertex as *lonely* if it has at most one big neighbour. Consider a minimum decent set D^* that contains the minimum number of lonely vertices.

Let v be a big vertex and suppose v is adjacent to t_v lonely vertices. If $t_v \geq 2$, then D^* contains exactly $t_v - 2$ lonely neighbours of v and all other small neighbours of v ; if $t_v \leq 1$, then D^* contains no lonely neighbour of v and all but at most $2 - t_v$ small neighbours of v . Hence, we know the number of lonely vertices in D^* ; so to determine D^* it suffices to determine the non-lonely vertices in D^* .

Consider an auxiliary multigraph H defined as follows. The vertex set of H is the big vertices of G . Two vertices u and v in H have k edges between them where k is the number of common non-lonely neighbours that u and v have. Choosing the non-lonely small vertices not in D^* is equivalent to choosing edges in H . Each vertex of H can have a maximum of 0, 1 or 2 incident

edges to be chosen, depending on the value of t_v . Specically, for $v \in V(H)$, let $f(v) = \max(0, 2 - t_v)$.

So the problem reduces to the f -factor problem on H : choose a maximum set of edges of H with upper bound $f(v)$ on the degrees at each vertex v . By a result of Tutte, this can be solved in polynomial time (see [16, pp. 125–126]). Thus we can determine D^* in polynomial time.

Now, define

$$f(G) = l(G) + d(G).$$

This is the smallest amount by which the degree-sum of G must be increased to eliminate all end-vertices and to make the empty set decent.

Lemma 4 *For all graphs G ,*

$$aug(G) \geq f(G)/2.$$

PROOF. The addition of an edge can decrease the value of f by at most 2. For, the addition of the edge increments the degrees of two vertices. This increment can convert an end-vertex into a vertex of degree 2, or a small vertex into a big vertex, but not both. Hence if the addition of an edge destroys two end-vertices it does not change the decent sets; if it destroys one end-vertex it can reduce the cardinality of a smallest decent set by at most 1; and if it does not destroy any end-vertex it can reduce the cardinality of a smallest decent set by at most 2.

Since, by Lemma 2, the augmented graph $A(G)$ with property 3VC satisfies $f(A(G)) = 0$, the result follows.

In this paper we solve the augmentation problem for trees and the 3VC property:

Theorem 5 *If T is a tree, then $aug(T) = \lceil f(T)/2 \rceil$.*

The proof of the upper bound is given in the next section. We discuss here some consequences.

Consider for example the path. If $G = P_n$ for $n \geq 3$, then $l(G) = 2$ and $d(G) = 0$. Certainly $aug(G) = 1$. Consider also the star with s leaves. If $G = K(1, s)$, then $l(G) = s$ and $d(G) = s - 2$. Thus $f(G)/2 = s - 1$. In fact,

by adding $s - 1$ edges such that the graph induced by the end-vertices is a path one obtains a hamiltonian graph.

This equality, however, does not hold in general. For example, for $m \geq 2$ the graph $G = K_2 + 3K_m$ has $f(G) = 0$ but does not satisfy the 3VC property.

3 Proof of Theorem 5

We now prove the upper bound of Theorem 5 for $aug(T)$. The proof is in three parts:

- (1) Reduce to the case where $d(T) = 0$;
- (2) Reduce to the case where $d(T) = 0$ and $l(T)$ is even; and
- (3) Solve the case where $d(T) = 0$ and $l(T)$ is even.

We start with part (3).

3.1 Compliant Trees

We say that a tree is *compliant* if it has an even number of leaves and every vertex is adjacent to at most two small vertices. We say that a big vertex is *critical* if it is adjacent to two small vertices.

Let T be a compliant tree with $2k$ leaves. We show that it is possible to augment T using k edges to have the property that every 3 vertices on a cycle. That is,

it is possible to add edges pairing off the end-vertices of T such that the resulting graph satisfies the 3VC property.

The proof is by induction on the number of vertices. Note that the base case follows since P_3 is the compliant tree of minimum order.

3.1.1 Penults

We define an r -penult as follows. If we root the tree at the central vertex/vertices and orient the edges away from the centre (and omit the central edge if any), then a vertex v is an r -penult if the longest path emanating from v has length r . For example, an 0-penult is an end-vertex, and a 1-penult is a vertex which is not an end-vertex but at most one neighbour is not an

end-vertex. Also, a vertex at distance r from the end in a diametrical path in a tree of diameter at least $2r$ is an r -penult.

We will examine the structure of subtrees rooted at penults. We need some lemmas. Recall that we are proceeding by induction; these lemmas show cases for which the induction is immediate.

Lemma 6 *We may assume: There is no edge $e = uv$ joining two small vertices.*

PROOF. Else contract the edge e by merging vertices u and v to a new (small) vertex x , augment the resulting tree T' to a graph $A(T')$ with the 3VC property, and then expand e again. (Any cycle containing x in $A(T')$ uses both edges incident with x and expands to a cycle containing e .)

From this it follows that 1-penults are big. In fact:

Corollary 7 *We may assume: a 1-penult has degree 3 with two end-vertex neighbours.*

Lemma 8 *We may assume: If edge $e = uv$ joins two big vertices, then at least one end must have both degree 3 and a critical neighbour other than u or v .*

PROOF. Remove the edge e and add one edge joining an end-vertex from each component. If the resulting tree T' is compliant, then, by Lemma 6, we can augment T' by pairing off end-vertices so that the resulting graph has the 3VC property. Hence T' is not compliant and thus one of the ends of e , say u , must become small when e is removed, and u must have a neighbour y in $T - e$ which now has three small neighbours. Note that $y \neq v$ and that y is critical in T .

Lemma 9 *We may assume: If edge $e = u_1u_2$ joins two big vertices neither of which is a 1-penult, then both components of $T - e$ must have an odd number of leaves.*

PROOF. Remove edge e and add vertex w_1 adjacent only to u_1 and vertex w_2 adjacent only to u_2 . Then add end-vertices x_1 and y_1 adjacent to w_1 , and end-vertices x_2 and y_2 adjacent to w_2 . Call the set of six added vertices D . The result is components T_1 and T_2 containing u_1 and u_2 respectively.

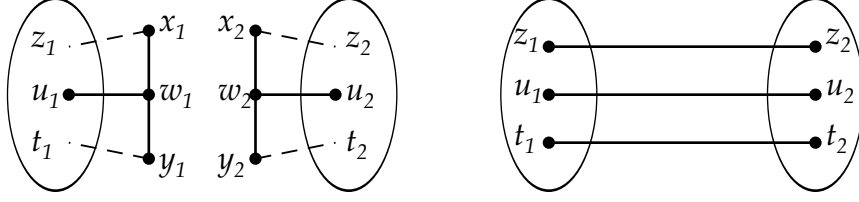


Fig. 2. Reduction in Lemma 9

If both components of $T - e$ have an even number of leaves, then both trees T_1 and T_2 are compliant. Furthermore, since neither end of e was a 1-penult, it follows that T_1 and T_2 have smaller order than T . Therefore by the inductive hypothesis we can, by pairing off end-vertices, augment T_1 to $A(T_1)$ and T_2 to $A(T_2)$ such that $A(T_1)$ and $A(T_2)$ satisfy 3VC. See Figure 2.

Assume that the following edges are added in the augmentation: x_1z_1, y_1t_1, x_2z_2 and y_2t_2 . (Note that in the augmentation x_1 is not paired with y_1 , since otherwise x_1, y_1 and u_1 do not lie on a common cycle; similarly x_2 is not paired with y_2 .) Now to create the augmentation $A(T)$, take the augmentations $A(T_1)$ and $A(T_2)$, delete the vertices of D , reinsert edge e and add the edges t_1t_2 and z_1z_2 .

We now argue that the result $A(T)$ satisfies 3VC. Consider any triple $S = \{a, b, c\}$ of vertices of T . There are two cases:

- S is contained in one of the components of $T - e$. Say $\{a, b, c\} \subseteq V(T_1)$. There exists a cycle C_1 in $A(T_1)$ through the vertices of S . If this cycle does not use w_1 then C_1 is a subgraph of $A(T)$. So suppose the cycle does use the vertex w_1 . If the two neighbours of w_1 in C_1 are x_1 and y_1 , then replace the z_1-t_1 path of C_1 with the edge z_1z_2 followed by the z_2-t_2 path in T_2 and then the edge t_2t_1 to produce a cycle in T containing S . If the two neighbours of w_1 in C_1 are u_1 and x_1 , then replace the u_1-z_1 path of C_1 with the edge u_1u_2 followed by the u_2-z_2 path in T_2 and then the edge z_2z_1 . Proceed in a similar manner if the two neighbours of w_1 in C_1 are u_1 and y_1 .
- S is not contained in one of the components of $T - e$. Say $\{a, b\} \subseteq V(T_1)$ and $c \in V(T_2)$. If there is a cycle C_1 in $A(T_1)$ through a, b, x_1, y_1 , then let C_2 be the cycle in $A(T_2)$ through c, x_2, y_2 . Combine C_1 and C_2 with edges t_1t_2 and z_1z_2 and delete D to produce the desired cycle.

Otherwise, in $A(T_1)$ there are cycles C_1^x through a, b, x_1 and C_1^y through a, b, y_1 that both use the edge u_1w_1 . Consider cycle C_2 in $A(T_2)$ through c, y_2, u_2 . If this cycle uses the edge u_2w_2 , then combine it with C_1^y using edges e and t_1t_2 and delete D . Otherwise there is a path P in T_2 from t_2 to z_2 that includes the vertices c and u_2 . If c occurs on the t_2-u_2 segment, then combine the t_2-u_2 segment of P with C_1^y , add e and t_1t_2 and delete D . Otherwise, combine the u_2-z_2 segment of P with C_1^x and edges z_1z_2 and e and delete D .

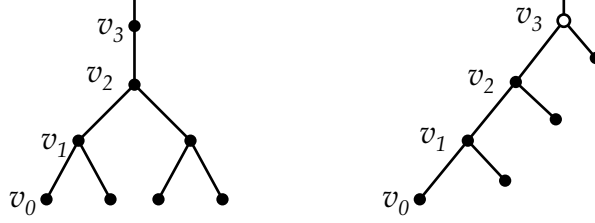


Fig. 3. The two types of 3-penults

Thus we may assume that T_1 and T_2 are not compliant.

We are now able to establish the structure of 3-penults:

Lemma 10 *We may assume: There exists a 3-penult. For any 3-penult v_3 , the subtree rooted at v_3 has one of the two structures shown in Figure 3. Note that v_3 in the second structure can have degree bigger than 3.*

PROOF. By Corollary 7, any 1-penult has degree 3. If there is no 2-penult in T , then the tree has diameter at most 3 and the result is easily checked. So we may assume that there exists a 2-penult. Consider any 2-penult v_2 with a 1-penult neighbour v_1 . Since v_1 has only v_2 as a critical neighbour, by Lemma 8, v_2 has degree 3 and a critical neighbour distinct from v_1 . If there is no 3-penult, then the tree has diameter at most 5 and the result is easily checked since interior vertices have degree 3. So we may assume that there exists a 3-penult.

Let v_i be i -penults for $0 \leq i \leq 3$ such that $v_3v_2v_1v_0$ is a path. As above, v_1 and v_2 both have degree 3. If the third neighbour of v_2 is an end-vertex, then v_3 is the critical neighbour of v_2 distinct from v_1 ; thus v_3 has two small neighbours. Since 1- and 2-penults are big, v_3 must therefore have at least one end-vertex neighbour. On the other hand, if the third neighbour of v_2 is a 1-penult, then the subtree descending from v_2 has four leaves. It follows from Lemma 9 that v_3 cannot be big, and therefore v_3 is small.

3.1.2 Two 3-penults

Suppose there are at least two 3-penults. By Lemma 10, there are three possibilities. The proof in the three cases is similar.

Case 1: *Two small 3-penults.* Suppose l_3 and r_3 are small 3-penults. Assume $l_3l_2l_1l_0$ is a path with l_i an i -penult ($0 \leq i \leq 3$). Similarly let $r_3r_2r_1r_0$ be a path with r_i an i -penult. Let D_l consist of the four vertices in the subtree at l_3 except for l_0, l_1, l_2 and l_3 ; let D_r consist of the four vertices in the subtree

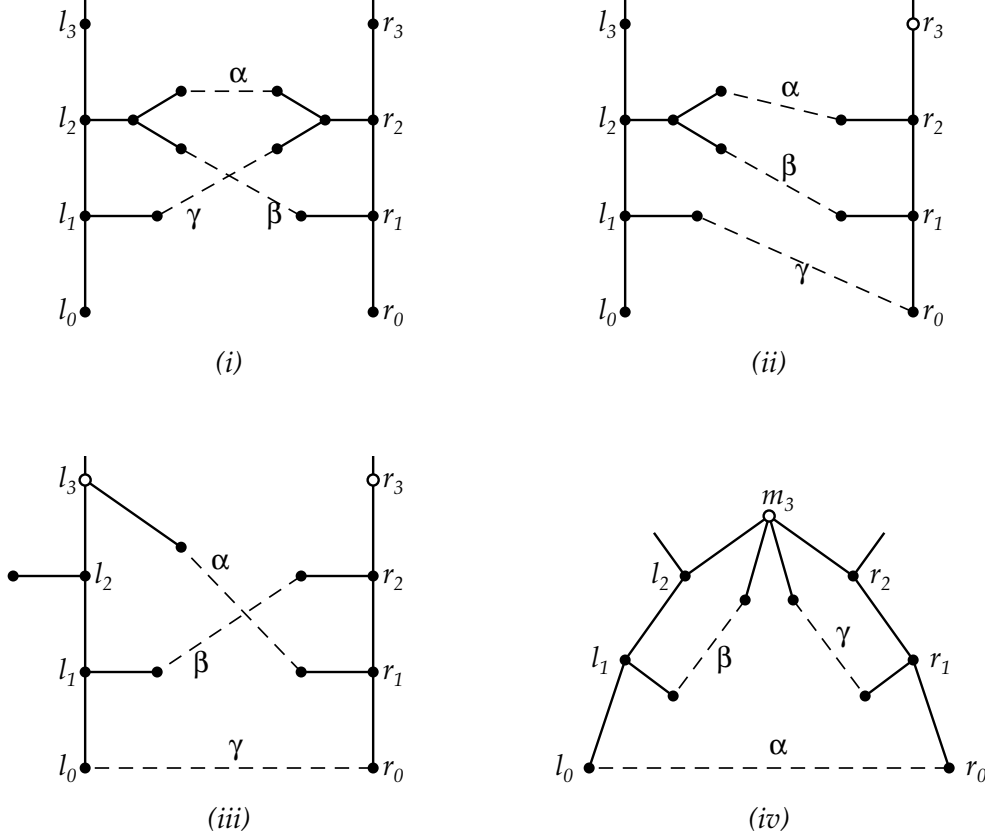


Fig. 4. Induction with penults

at r_3 except for r_0, r_1, r_2 and r_3 . Call $D = D_l \cup D_r$. The tree $T - D$ has six fewer leaves than T .

The tree $T - D$ is compliant since l_3 and r_3 are small in T . So by the inductive hypothesis we can augment $T - D$ using $k - 3$ edges to a graph $A(T - D)$ that satisfies the 3VC property.

To find the augmentation $A(T)$ of T , re-insert the vertices of D and add three edges each joining an end-vertex in D_l with an end-vertex in D_r without creating a 6-cycle. There is only one way up to symmetry of doing this. Let α denote the edge joining a descendant of l_2 to one of r_2 , β the edge joining a descendant of l_2 to one of r_1 , and γ the edge joining a descendant of l_1 to one of r_2 . See Figure 4(i).

Consider a triple S of vertices of T . If $D \cap S = \emptyset$, then the cycle in $A(T - D)$ suffices. So consider the two cases:

- $|D \cap S| = 1$. If the vertex in $D \cap S$ is on α or β or is a 1-penult, then take a cycle C in $A(T - D)$ through the vertices of $S - D$ and r_3 . This cycle uses the edge $r_1 r_2$. Delete the edge $r_1 r_2$ and insert into C the $r_1 - r_2$ path that contains α and β to obtain the desired cycle. (Note that it does not matter

whether C uses l_3 or not.)

Similarly, if the vertex in $D \cap S$ is on γ , then take a cycle in $A(T - D)$ through $S - D$ and l_3 . Delete the edge $l_1 l_2$ and insert into C the l_1 - l_2 path that contains α and γ .

- $|D \cap S| \geq 2$. Find a cycle C in $A(T - D)$ through $S - D$, l_3 and r_3 . This cycle uses the vertices l_1 and r_1 . Consider the path P from r_1 to l_1 that uses β , α , γ . It uses all of D . Replace the l_1 - r_1 subpath of C that does not contain a vertex of S with P .

Case 2: *A small 3-penult and a big 3-penult.* Suppose l_3 is a small 3-penult and r_3 a big 3-penult. Let $l_3 l_2 l_1 l_0$ and $r_3 r_2 r_1 r_0$ be paths of decreasing penults as before. Let D_l consist of the four vertices in the subtree at l_3 except for l_0 , l_1 , l_2 and l_3 ; let D_r consist of the five vertices r_2 and r_1 and their descendants. Call $D = D_l \cup D_r$. The tree $T - D$ has six fewer leaves than T .

The tree $T - D$ is compliant. To see this, observe that l_3 is small, and that since r_3 is critical if r_3 is small in $T - D$ both its neighbours are small. So by the inductive hypothesis we can augment $T - D$ using $k - 3$ edges to a graph $A(T - D)$ that satisfies the 3VC property.

To find the augmentation $A(T)$ of T , re-insert the vertices of D and add three edges each joining an end-vertex in D_l with an end-vertex in D_r without creating a 6-cycle. Let α denote the edge joining a descendant of l_2 to one of r_2 , β the edge joining a descendant of l_2 to one of r_1 , and γ the edge joining r_0 to the descendant of l_1 . See Figure 4(ii).

Consider a triple S of vertices of T . If $D \cap S = \emptyset$, then the cycle in $A(T - D)$ suffices. So consider the two cases:

- $|D \cap S| = 1$. Take a cycle C in $A(T - D)$ through the vertices of $S - D$ and l_0 . Insert into C either the l_2 - l_1 path that uses α and γ or the one that uses β and γ . (Note that C uses none of r_0 , r_1 or r_2 .)
- $|D \cap S| \geq 2$. Find a cycle C in $A(T - D)$ through $S - D$, l_0 and r_3 and consider the path P from r_3 to l_1 that uses α , β and γ , and proceed as in Case 1.

Case 3: *Two big 3-penults.* Suppose l_3 and r_3 are big 3-penults. Let $l_3 l_2 l_1 l_0$ and $r_3 r_2 r_1 r_0$ be paths of decreasing penults. Let D_l consist of the four vertices l_1 , its end-vertex neighbours and an end-vertex neighbour of l_3 ; let D_r consist of r_2 and its four descendants. Call $D = D_l \cup D_r$.

The tree $T - D$ is compliant since l_3 and r_3 are critical in T and so if either vertex is small in $T - D$ both its neighbours in $T - D$ are small. Thus, by the inductive hypothesis we can augment $T - D$ using $k - 3$ edges to a graph $A(T - D)$ that satisfies the 3VC property.

To find the augmentation $A(T)$ of T , re-insert the vertices of D and add three edges α, β, γ each joining an end-vertex in D_l with an end-vertex in D_r without creating a 6-cycle. See Figure 4(iii).

Consider a triple S of vertices of T . If $D \cap S = \emptyset$, then the cycle in $A(T - D)$ suffices. So consider the two cases:

- $|D \cap S| = 1$. Take a cycle C in $A(T - D)$ through the vertices of $S - D$ and l_2 . Insert into C either the l_3 - l_2 path that uses α and β or the one that that uses α and γ .
- $|D \cap S| \geq 2$. Find a cycle C in $A(T - D)$ through $S - D$, l_3 and r_3 and consider the path P from l_3 to r_3 that uses α, γ and β , and proceed as in Case 1.

3.1.3 Unique 3-penult

Finally, we consider the case where there is only one 3-penult; call it m_3 . Since we have already dealt with the case where the diameter is at most 5, we may assume that the graph has diameter 6. Thus this 3-penult has two disjoint subtrees each of one of the types described above.

There are two cases to consider. If m_3 is small, then proceed as in Case 1 above: the fact that $l_3 \neq r_3$ is not used in Case 1.

Assume m_3 is big. Then the two subtrees of m_3 are as in the second structure of Figure 3. Since a vertex of degree 2 has to be a k -penult for k at least 3, the two small neighbours of m_3 are both end-vertices. Let l' and r' be the end-vertices adjacent to m_3 and assume $m_3 l_2 l_1 l_0$ and $m_3 r_2 r_1 r_0$ are paths of decreasing penults. Let D consist of l_1 and its descendants, r_1 and its descendants, as well as l' and r' .

The tree $T - D$ is compliant since m_3 loses two small neighbours and gains two small neighbours. So by the inductive hypothesis we can augment $T - D$ using $k - 3$ edges to a graph $A(T - D)$ that satisfies the 3VC property.

To find the augmentation $A(T)$ of T , re-insert the deleted vertices D and add three edges each joining two different end-vertices in D as follows: α joins l_0 and r_0 , β joins l' to the other descendant of l_1 and γ joins r' to the other descendant of r_1 . See Figure 4(iv).

Consider a triple S of vertices of T . If $D \cap S = \emptyset$, then the cycle in $A(T - D)$ suffices. So consider the three cases:

- $|D \cap S| = 1$. Then take a cycle C in $A(T - D)$ through the vertices of $S - D$ and l_2 , and insert the m_3 - l_2 path using the vertex of $D \cap S$.

- $|D \cap S| = 2$ and $m_3 \notin S$. Then find a cycle C in $A(T - D)$ through $S - D$, l_2 and r_2 . This cycle must include m_3 . Suppose S includes an end of β and an end of γ . Then let P be the l_2 - r_2 path that uses m_3 and all of $D \cap S$. Delete m_3 from C and insert P to obtain the desired cycle. Otherwise let P be the m_3 - l_2 or m_3 - r_2 path that uses all of $D \cap S$. Insert P into C as in Case 1 above.
- $|D \cap S| = 3$, or $|D \cap S| = 2$ and $m_3 \in S$. The vertices $D \cup \{m_3\}$ induce a cycle.

This completes the proof of the case $d(T) = 0$ and $l(T)$ even.

3.2 Reduction

We start with some preliminaries.

Lemma 11 *Suppose G' is obtained from G by the addition of $(f(G) + \varepsilon)/2$ edges and that $f(G') = 0$. Then the set X of vertices that are big in G' but small in G has cardinality at most $d(G) + \varepsilon$. In particular, if $\varepsilon = 0$ then X must be a minimum decent set of G .*

PROOF. Let x denote the number of vertices that are big in G' but small in G . Then, since $f(G') = 0$,

$$\sum_v \deg_{G'}(v) \geq \sum_v \deg_G(v) + l(G) + x,$$

while we know that

$$\sum_v \deg_{G'}(v) = \sum_v \deg_G(v) + f(G) + \varepsilon,$$

Thus $x \leq f(G) - l(G) + \varepsilon = d(G) + \varepsilon$, as required.

Lemma 12 *If G satisfies 3VC and x is a vertex of degree 2 with neighbours y and z , then G' formed by contracting out x (deleting x and adding edge yz) also satisfies 3VC.*

PROOF. Consider any triple W of $V(G')$. Then there is a cycle C of G which includes all of W . If this cycle uses x , then it must use the segment yxz and thus the contracting out of x results in a new cycle which includes all of W . Note that the observation still follows if y and z are adjacent, but in this case the edge yz need not be duplicated.

3.2.1 Reduction to the case $l(T)$ even

Recall that we are still considering the case $d(T) = 0$ and suppose $l(T)$ is odd.

Let M be the number of edges joining small vertices to big vertices and N the number of edges joining two small vertices. Then

$$M + 2N = \sum_{v \text{ small}} \deg v = 2k + l$$

where k is the number of vertices of degree 2. Since l is odd, it follows that M is odd. Thus there exists a big vertex v with an odd number of small neighbours. Since $d(T) = 0$, v has exactly one small neighbour. Add a new end-vertex x adjacent only to v to form tree T' .

Then $l(T') = l(T) + 1$, and $d(T') = d(T) = 0$. Since the upper bound is true for l even, we can augment T' with $(l(T) + 1)/2$ edges to a graph $A(T')$ with the 3VC property. This augmentation $A(T')$ is obtained by adding edges joining pairs of end-vertices, and so x has degree 2 in T' . Thus by Lemma 12 we can form the desired augmentation of T by contracting out x .

3.2.2 Reduction to the case $d(T) = 0$

We now show that:

Since the upper bound $aug(T) \leq \lceil f(T)/2 \rceil$ is true for trees T with $d(T) = 0$, the upper bound is true for all trees.

The proof is by induction on $d(T)$. The base case is proved in subsection 3.2.1 above.

Let T be a tree with $d(T) > 0$. Then there exists a vertex v which is adjacent to at least three small vertices. There are two cases to consider.

Case 1: v is not adjacent to any vertex of degree 2. Let y be any end-vertex adjacent to v . Define $T' = T - y$. Clearly, $l(T') = l(T) - 1$.

We argue that $d(T') = d(T) - 1$. Note that if v is small in T' then T is $K(1, 3)$ and the claim still holds; so we may assume that v is not small in T' . Since we can add y to any decent set of T' and obtain a decent set of T , it follows that $d(T') \geq d(T) - 1$. On the other hand, let D^* be a smallest decent set of T . Then it must include one of the end-vertices adjacent to v , so we may assume that $y \in D^*$. The set $D^* - \{y\}$ is decent in T' ; this proves the reverse inequality and hence the claim. Thus, $f(T') = f(T) - 2$.

By the inductive hypothesis, T' can be augmented to a graph $A(T')$ which satisfies the 3VC property by adding $\lceil f(T')/2 \rceil$ edges.

Since v has two end-vertices as neighbours in T' , it follows from Lemma 11 that at least one neighbour of v is small in $A(T')$. Suppose s is a neighbour of v that is small in $A(T')$. Then form $A(T)$ by adding vertex y and making it adjacent to v and to s . The graph $A(T)$ is a supergraph of T with $\lceil f(T)/2 \rceil$ additional edges.

It remains to show that $A(T)$ satisfies the 3VC property. Consider any triple W of vertices of T . If $y \notin W$ then there is a cycle of $A(T')$ that includes all of W , and $A(T)$ contains $A(T')$. So assume that $y \in W$. Then let $W' = (W - \{y\}) + \{s\}$. By the construction there exists a cycle C' including W' in $A(T')$. Since s has degree 2 in $A(T')$, the cycle C' must use the edge sv . If we take C' and replace the edge sv by the path syv , we obtain a cycle through W in G . Hence G satisfies 3VC.

Case 2: v is adjacent to a vertex of degree 2. Then there exists a minimum decent set D^* and a degree-2 neighbour y of v such that $y \in D^*$. To see this, note that at least one small neighbour x of v is in D^* : if this x is an end-vertex, and y is a degree-2 neighbour of v , then we may replace x with y in D^* to obtain the desired decent set.

This case has four subcases. The method is the same in each: add vertices to T to form tree T' with a reduced value of d but with $\lceil f(T')/2 \rceil = \lceil f(T)/2 \rceil$, augment T' with $\lceil f(T')/2 \rceil = \lceil f(T)/2 \rceil$ edges to produce a graph with the 3VC property, and contract out the added vertices.

2.1 $f(T)$ even. Then form T' from T by adding a new vertex u that is adjacent only to y . Clearly, $l(T') = l(T) + 1$.

We claim that $d(T') = d(T) - 1$. Since y is no longer small, and y has at most two small neighbours, the set $D^* - \{y\}$ is decent in T' . Hence $d(T') \leq d(T) - 1$. Conversely, take any minimum decent set in T' and add y ; this creates a set that is decent in T . Hence $d(T) \leq d(T') + 1$. This establishes the claim, and so $f(T') = f(T)$.

By induction, there exists an augmentation $A(T')$ of T' with $f(T')/2$ additional edges and the 3VC property. By Lemma 11, u must have degree 2 in $A(T')$. So form $A(T)$ by contracting out u . This is an augmentation of T with $f(T')/2 = f(T)/2$ additional edges. By Lemma 12, $A(T)$ satisfies 3VC.

2.2 $f(T)$ is odd and y 's other neighbour w is big. Then form T' from T by adding two new vertices u and u' that are each only adjacent to y . Clearly, $l(T') = l(T) + 2$, and as in Case 2.1 $d(T') = d(T) - 1$, so that $f(T') = f(T) + 1$.

By induction, there exists an augmentation $A(T')$ of T' with $f(T')/2$ additional edges where $A(T')$ satisfies 3VC. By Lemma 11, in $A(T')$ both u

and u' have degree 2. Then form $A(T)$ from $A(T')$ by contracting out both u and u' in succession. (Vertices u and u' are nonadjacent since $A(T')$ is 2-connected.)

- 2.3 $f(T)$ is odd and y 's other neighbour w has degree 1. Then form T' from T by adding a new vertex u that is adjacent only to y . As above, $l(T') = l(T) + 1$ and $d(T') = d(T) - 1$. So $f(T') = f(T)$.

By induction, there exists an augmentation $A(T')$ of T' with $(f(T') + 1)/2$ additional edges where $A(T')$ satisfies 3VC. From Lemma 11, it follows that in $A(T')$ at least one of u or w has degree 2. We may assume that this is u . Then form $A(T)$ from $A(T')$ by contracting out u , and conclude as above.

- 2.4 $f(T)$ is odd and y 's other neighbour w has degree 2. Here there are two subcases.

a) If there is some choice of D^* that contains w , then proceed as follows. Form T' by adding two vertices u and u' that are both adjacent only to y .

Clearly, $l(T') = l(T) + 2$. Since w is in $D^* - \{y\}$, that set is decent in T' and hence $d(T') \leq d(T) - 1$. Conversely, take any minimum decent set in T' : it must contain one of the small neighbours of y and we may assume that this is w ; so the addition of y creates a set that is decent in T and hence $d(T) \leq d(T') + 1$. Thus $f(T') = f(T) + 1$.

By induction, there exists an augmentation $A(T')$ of T' with $f(T')/2$ additional edges where $A(T')$ satisfies 3VC. By Lemma 11, in $A(T')$ both u and u' have degree 2. Then form $A(T)$ from $A(T')$ by contracting out both u and u' and conclude as above.

b) If no choice of D^* contains w , then proceed as follows. (Note that this means that no minimum decent set contains both w and y .) Form T' by introducing two new vertices u and u' and making u adjacent only to y and u' only adjacent to w .

Clearly $l(T') = l(T) + 2$. Since both y and w have at most two small neighbours in T' , it follows that $D^* - \{y\}$ is decent in T' . Hence $d(T') \leq d(T) - 1$. Conversely, suppose that $d(T') < d(T) - 1$. Then take a minimum decent set of T' and add y and w . The result must be decent in T . But since it has cardinality $d(T)$ we have a contradiction. Hence $d(T') \geq d(T) - 1$. Thus $f(T') = f(T) + 1$.

By induction, there exists an augmentation $A(T')$ of T' with $f(T')/2$ additional edges where $A(T')$ satisfies 3VC. From Lemma 11, in $A(T')$ both new vertices u and u' have degree 2. Then form $A(T)$ from $A(T')$ by contracting out both u and u' and conclude as above.

This completes the proof of Theorem 5.

We note that the proof provides for a polynomial-time algorithm for finding the minimum set of edges. In fact, if n is the order of the tree, the algorithm can be implemented in $O(n^2)$ time. In the case of compliant trees, the proof in general uses a reduction to a smaller tree, except in Lemma 9 where the reduction is to two smaller trees whose orders sum to 6 more than the original order. Even here the worst case is one tree of order $n - 1$ and one tree of order 7 (and the latter takes constant time to augment). It takes linear time to recognise which reduction is applicable, to modify the tree, and to modify the resultant augmentation. Hence we have the following recurrence for $t(n)$ the time taken for a compliant tree on n vertices:

$$t(n) = t(n - 1) + O(n).$$

This implies that $t(n)$ is $O(n^2)$. It follows immediately from Section 3.2.1 that all trees with $d = 0$ can be handled in $O(n^2)$ time. For general trees, the proof uses a reduction which reduces d , though the order may grow by 4. This reduction is undergone at most n times, and the resultant compliant tree has order at most $5n$. Since the recognition, reduction, and augmentation adjustment again take linear time, this shows that the overall algorithm also takes $O(n^2)$ time.

4 Open Problems

Several augmentation problems remain unsolved. For example the problem of augmenting a tree T so that every k vertices lie on cycle, for $|V(T)| > k \geq 4$. Of course the problem of augmenting a general graph to have the 3VC property is still open, as is the problem of augmenting a tree to be k -connected.

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